

Yizheng Xie

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Education

Brown University , Providence, RI, USA Ph.D. in Computer Science (Distributed System, Operating System, Cloud Computing)	Sept 2023 – Expected May 2028
Boston University , Boston, MA, USA M.S. in Computer Science	Sept 2021 – May 2023
Chinese University of Hong Kong, Shenzhen , Shenzhen, Guangdong, China B.E. in Electronic Information Engineering	Sept 2017 – May 2021

Research

Slowpoke: End-to-end Throughput Optimization Modeling for Microservices <i>To appear at NSDI'26, Microservice, Tracing, Kubernetes, Service Mesh, Docker, Golang, C++</i>	Sept 2023 – Apr 2025
- Propose a novel linear programming-based performance model to quantify end-to-end throughput improvements of hypothetical microservice optimizations without requiring white-box knowledge. - Develop the first system to accurately predict throughput in complex microservices, achieving RMSE of 2.1% across 4 real-world benchmarks and 1.9% across 108 synthetic benchmarks with diverse characteristics. - Design a lightweight distributed slowdown mechanism enabling coordinated pausing across large-scale microservices.	
λEASH: Scaling Out Shell Scripts with Recoverable Serverless Computing <i>Under submission of EuroSys'26, Serverless, AWS Lambda, Recovery, Shell, Python, C, Rust</i>	Sept 2023 – Jan 2025
- Develop a system to scale out unmodified shell scripts, achieving $15.2\times$ average and $215.8\times$ peak speedups over Bash with minimal cost overhead; in some cases both faster and cheaper. - Design a recoverability protocol that extends AWS Lambda beyond the 15-minute execution limit while preserving correctness and Unix-style streaming efficiency, with negligible slowdown (avg: $0.03\times$). - Propose a hybrid scheduler to balance workloads across EC2 and Lambda, enabling direct TCP communication between Lambdas via hole-punching and automatically resolving software dependencies despite bandwidth constraints.	
Fractal: Fault-Tolerant Shell-Script Distribution <i>To appear at NSDI'26, Fault-Tolerance, Hadoop Distributed File System, Shell, Golang, C</i>	Sept 2023 – Dec 2024
- Develop a system for fault-tolerant distribution of unmodified shell scripts, delivering average speedups of $9.6\times$ over Bash, $5.5\times$ over Hadoop Streaming (AHS), and 17% over DISH, the state-of-the-art shell scaleout system. - Design a lightweight fault-recovery mechanism—including remote-pipe instrumentation and dynamic result persistence—that achieves recovery within $1.26\times$ of fault-free execution and delivers a $9.3\times$ speedup over AHS. - Introduce a large-scale fault-injection tool to precisely characterize recovery behavior under real-world conditions.	

Working Experience

Meta Platforms , Menlo Park, CA <i>Production Engineer Intern in Kernel Team (Python, Fedora, Rust Packaging, Upstream Userspace)</i>	May 2022 – Aug 2022
- Integrate Fedora and CentOS Stream forges and issue trackers (Pagure, GitLab, Bugzilla) into Meta Internal Task tool. - Improve open-source tools, e.g., fixing Bugzilla authentication issues and adding Copr build integration in review tool. - Implement a CLI tool to automate the process of updating Rust RPM packages by recursively updating dependencies and packaging missing dependencies, analyzing parallel chain-build order and checking compatibility for the current update set or any Bodhi rust update set. (Pagure Repository: pagure.io/fedora-rust/rust-update-set)	

Publications

- [1] Zhicheng Huang, Ramiz Dunder, Yizheng Xie, Konstantinos Kallas, and Nikos Vasilakis. Fractal: Fault-tolerant shell-script distribution. In *23rd USENIX Symposium on Networked Systems Design and Implementation*, Renton, WA, May 2026. USENIX Association.
- [2] Yizheng Xie, Di Jin, and Nikos Vasilakis. Towards hybrid cooperative-preemptive scheduling. In *Proceedings of the 13th Workshop on Programming Languages and Operating Systems*, Seoul, Republic of Korea, October 2025. ACM.
- [3] Yizheng Xie, Di Jin, Oğuzhan Çölkesen, Vasiliki Kalavri, John Liagouris, and Nikos Vasilakis. Slowpoke: End-to-end throughput optimization modeling for microservice applications. In *23rd USENIX Symposium on Networked Systems Design and Implementation*, Renton, WA, May 2026. USENIX Association.